# Lennox Sama Keeble

Curriculum Vitae

Last updated: October 16, 2026.

## Education

**Wake Forest University** **2026 - 2031**
*Doctor of Philosophy (Ph.D.) in Physics. Supervisor: Professor Alejandro Cárdenas-Avendaño.* *Winston-Salem, USA*

**University of Cambridge** **2025 - 2026**
*Master of Advanced Study (MASt) in Theoretical Physics (Part III of the Mathematical Tripos).* *Cambridge, UK*
*Supervisor: Dr. Lorenzo Gavassino.*

**Princeton University** **2020 - 2025**
*Bachelor of Arts (B.A.) in Physics. Supervisor: Professor Frans Pretorius.* *Princeton, USA*

## Publication Summary

- **8** (submitted/accepted) refereed papers; **6** as (co-)lead author.
- **Papers by research area:** relativistic hydrodynamics (**3**); black hole imaging (**2**); extreme mass-ratio inspirals (**2**); black hole perturbation theory (**1**).
- Links to academic metrics: Google Scholar, INSPIRE.

## Collaboration Memberships

- Laser Interferometer Space Antenna (LISA) Consortium Core Member (2026 –).
- Simons Collaboration on Black Holes and Strong Gravity (2026 –).

## Honours, Scholarships & Awards

- Cambridge Trust Master's Studentship, University of Cambridge, 2025.
- Academic Futures Graduate Scholarship, University of Oxford, 2025. (Declined.)
- Allen G. Shenstone Prize in Physics, Princeton University, Department of Physics, 2025.
- Manfred Pyka Memorial Prize, Princeton University, Department of Physics, 2022.

## Service

- Refereeing activities for Physical Review D (PRD).

## Invited Talks, Conferences & Workshops

Bold **numbers** indicate a contributed talk; bold **text** indicates an invited talk.

8. IGWN-PI School on Gravitational Waves, Perimeter Institute for Theoretical Physics, 2026.
7. Relativistic Astrophysics Summer School, Caltech, 2025.
**6.** Nuclear Physics of MultiMessenger Mergers (NP3M) Summer School, Indiana University, 2025.
**5. Los Alamos Astrophysics Distinguished Seminar Series, Center for Nonlinear Studies, Los Alamos National Laboratory, 2025.**
4. Black Hole Mimickers: From Theory to Observation, Princeton Center for Theoretical Science & Princeton Gravity Initiative, 2025.
3. $CU^2MIP$ Undergraduate Conference, College Park, USA, 2024.
**2.** American Physical Society April Meeting, 2024.
1. Taking it to the extreme: Symmetries and dynamics of extremal black holes, Princeton Center for Theoretical Science & Princeton Gravity Initiative, 2024.

## Professional Experience

**Investment Associate Intern** — **Summer 2024**
*Bridgewater Associates* — *Westport, Connecticut, USA*

**Summer Business Analyst** — **Summer 2022**
*McKinsey & Company* — *San Francisco, California, USA*

## Miscellaneous

**Programming Languages:** julia., Python, Mathematica & Slurm.
**Web Links:** Webpage, GitHub.

## List of Publications

† indicates first-authored papers or alphabetical ordering.

### Refereed (submitted/accepted)

[8] *Breakdown of hydrodynamic universality in causal first-order viscous neutron star oscillations.*
† **L. S. Keeble** and A. Cárdenas-Avendaño.
arXiv: 2610.21966 (gr-qc).

[7] *Quadratic Gravitational-Wave Scattering by Kerr Black Holes.*
† **L. S. Keeble**, H. Zhu, L. E. Kidder, H. P. Pfeiffer, and M. A. Scheel.
arXiv: 2609.25522 (gr-qc).

[6] *A Systematic Study of Resonance-Driven Flux Modifications in Extreme-Mass-Ratio Inspirals.*
E. Levati, **L. S. Keeble**, and A. Cárdenas-Avendaño.
arXiv: 2609.25392 (gr-qc).

[5] *Radial oscillations of viscous stars.*
† **L. S. Keeble** and J. Redondo-Yuste.
Physical Review D 113, 124058 (2026) arXiv: 2603.23622 (gr-qc).

[4] *First-order viscous relativistic hydrodynamics on the two-sphere.*
† **L. S. Keeble** and F. Pretorius.
Physical Review D 112, 124034 (2025) arXiv: 2508.20998 (gr-qc).

[3] *Estimating high-order time derivatives of Kerr orbital functionals.*
† **L. S. Keeble** and A. Cárdenas-Avendaño.
Physical Review D 112, 084048 (2025) arXiv: 2508.08888 (gr-qc).

[2] *Inferring black hole spin from interferometric measurements of the first photon ring: A geometric approach.*
† **L. S. Keeble**, A. Cárdenas-Avendaño, and D. C. M. Palumbo.
Physical Review D 111, 103042 (2025) arXiv: 2502.20312 (astro-ph.HE).

[1] *Assessing the impact of instrument noise and astrophysical fluctuations on measurements of the first black hole photon ring.*
A. Cárdenas-Avendaño, **L. Keeble**, and A. Lupsasca.
Physical Review D 109, 124052 (2024) arXiv: 2404.01083 (gr-qc).

### White Papers and Conference Proceedings

[1] *The Black Hole Explorer: photon ring science, detection, and shape measurement.*
A. Lupsasca, A. Cárdenas-Avendaño, D. C. M. Palumbo, M. D. Johnson, S. E. Gralla, D. P. Marrone, P. Galison, P. Tiede, and **L. Keeble**.
Proc. SPIE Int. Soc. Opt. Eng. 13092, 130926Q (2024) arXiv: 2406.09498 (gr-qc).